MIL-I-25859B (USAF)
24 September 1971
SUPERSEDING
MIL-I-25859A (USAF)
31 December 1957

MILITARY SPECIFICATION

INDICATOR, THRUST AEU-2/A, SYNCHRO STYLE, PRESSURE RATIO, 1.2-3.4 UNITS, INTEGRALLY LIGHTED

1. SCOPE

1.1 SCOPE. This specification covers the requirements for one type of integrally lighted, synchro style, pressure ratio thrust indicator designated AEU-2/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For
MIL-I-25467	Lighting, Integral, Aircraft Instrument, General Specification For
MIL-T-25485	Transmitter, Thrust, Single Synchro Style, Pressure Ratio, Type MK-2
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby, Synthetic

STANDARDS

Federal

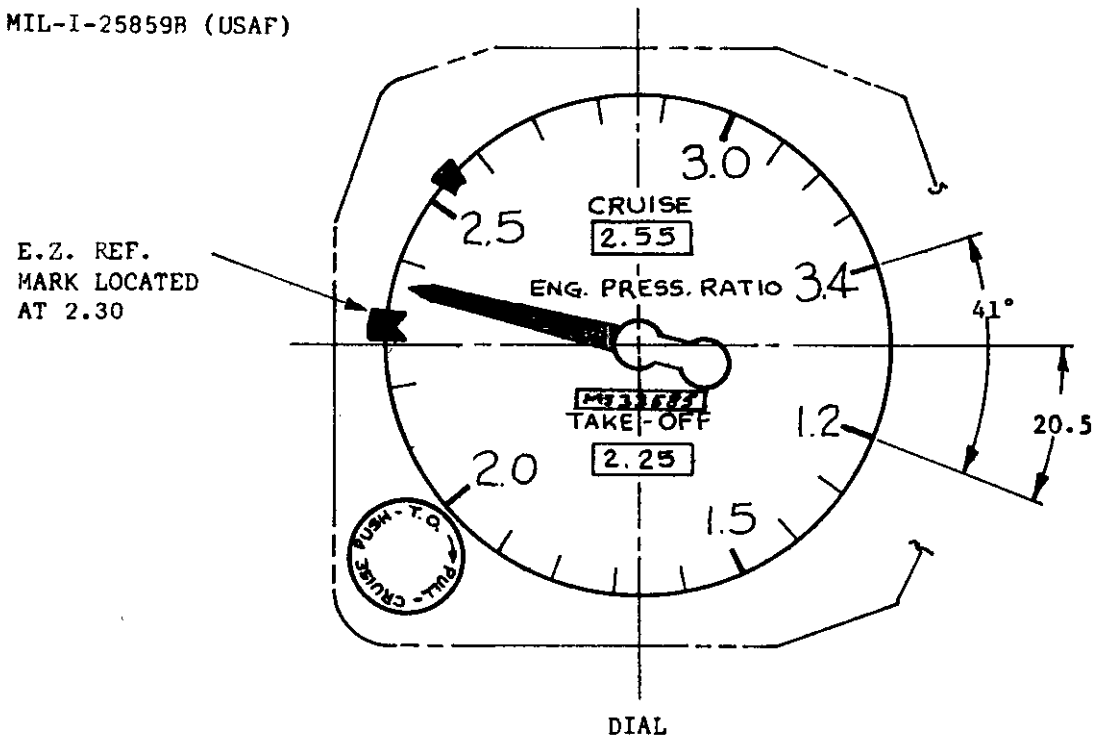
FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property

FSC: 6620

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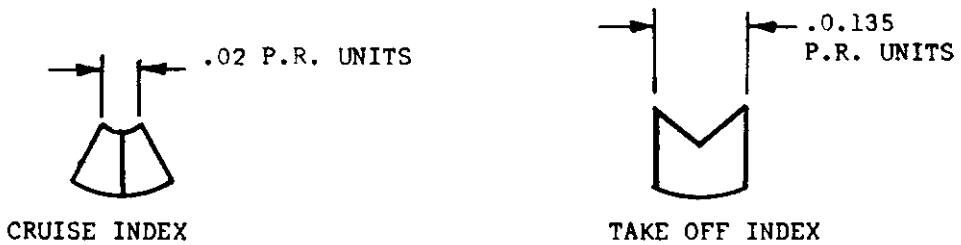


FIGURE 2 INDEX AND DIAL

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Marking	Height or Length Inch ± 1/62	Width of Line or Graduation Inch ± 0.005
Numerals 1.5, 2.0, 2.5, and 3.0	5/32	0.020
Numerals 1.2 and 3.4	1/8	0.016
Counter numerals	1/8	0.016
Major graduations	3/16	0.020
Minor graduations	5/32	0.016
Lettering ENG. PRESS. RATIO	3/32	0.016
Lettering CRUISE	1/8	0.016
Lettering TAKE-OFF	1/8	0.016
Electrical zero reference mark	5/32	0.016

3.3.3 Counters and indices.

3.3.3.1 Takeoff index and counter. The dial shall contain a moveable index and a drum-type counter for takeoff-pressure-ratio reference indication. The index shall conform to figure 2, and shall be mechanically related to the counter reading in pressure ratio as shown in figure 2. The maximum allowable position error between the counter and the center V of the index shall be ± 0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.20 to 3.40. The index and associated counter shall move smoothly over the complete scale range of 1.20 to 3.40 pressure ratio when a torque of 6 ± 3 inch ounces is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is pushed and turned. The length of the index shall be 0.156 + 0.000 or -.031 inch, depending on the pointer length (see 3.3.4). The counter numerals shall move downward with an increasing pressure ratio as the knob is rotated clockwise. The mechanism employed for driving the index and counter shall be rugged in design. The control knob and drive shaft shall conform to the dimensions specified in figure 1. There shall be no binding, deformation, or malfunction when the indexing mechanism is engaged and rotated over its design range. Suitable rugged stops or slip-clutch mechanism shall be provided to prevent the index and counter from being damaged or mismatched if rotated above or below the dial range.

3.3.3.2 Cruise index and counter. The dial shall contain a moveable index and a drum-type counter for accurate cruise-pressure-ratio reference indication. The index shall be mechanically related to the counter reading in pressure ratio as shown in figure 2. The maximum allowable position error between the counter and the center line of the index shall be ± 0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.20 to 3.40. The index and associated counter shall move smoothly over the complete scale range of 1.20 to 3.40 pressure ratio when a torque of 6 ± 3 inch ounces is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is pulled and turned. The length of the index shall be 0.156 + 0.000 or -0.031 inch, depending on the pointer length (see 3.3.4). The counter numerals shall move downward for increasing pressure ratio as the knob is rotated clockwise. The mechanism employed for

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4.6.2 Position error. The tolerance specified in Table II shall apply.

4.6.3 Vibration error. The tolerance specified in Table II shall apply. Upon completion of the vibration test, the test specified under 4.6.1 shall be repeated.

4.6.4 Low temperature. While at the test temperature, the indicator shall be tested for scale error and friction. Errors shall not exceed the respective tolerances specified in Tables I and II. The indicator shall not require more than 5 seconds to come within the specified tolerance.

4.6.5 High temperature. While at the test temperature, the indicator shall be tested for scale error and friction. Errors shall not exceed those specified in Tables I and II.

4.6.6 Thermal shock. The indicator shall be subjected to six cycles of immersion in water at a temperature of 85° ± 2° C and then in water at 5° ± 2°C without evidence of moisture penetration or damage to the coating or case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After six continuous cycles have been completed, the indicator shall be subjected to the sealing test of MIL-I-7057. Leakage in excess of that permitted or fogging of the inner side of the cover glass shall be cause for rejection.

4.6.7 Counter cycling. The control knob shall be pushed in and turned continuously in order to move the takeoff index and counter from 1.3 ± 0.1 to 3.3 ± 0.1 and return to 1.3 ± 0.1 pressure ratio units. The control knob shall then be pulled out and turned continuously in order to move the cruise index and counter 1.3 ± 0.1 to 3.3 ± 0.1 and return to 1.3 ± 0.1 pressure ratio units. This procedure shall be considered one cycle of operation. The total cycling period shall be a minimum of 2,500 cycles and shall be conducted at a cycling rate of 1 ± 0.2 cpm. The last 500 cycles shall be conducted over the entire range of the scale and shall cause the slip clutch to engage for at least 1 complete revolution of the set knob at both ends of the scale. If stops are used instead of slip clutches, the index shall be driven against the stop with a minimum torque of 24 ounce inches. Prior to the cycling test, the torque required to turn the control knob shall be measured and recorded. The allowable torque required to turn the control knob throughout the takeoff and cruise range of 1.2 to 3.4 pressure ratio shall not exceed 6 ± 3 ounce inches. Upon completion of the 2,500 cycle testing, physical and functional operation inspection shall be conducted to assure that binding, deformation, and malfunction is not evident. The torque required to turn the control knob throughout the takeoff and cruise range of 1.2 to 3.4 pressure ratio shall not exceed the measurements recorded at the start of the cycling test by more than 2 ounce inches.

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4.6.8 Low temperature exposure. While at the test temperature the scale error and friction shall not exceed the tolerances specified in Tables I and II. After the indicator has been maintained at room temperature for not less than 4 hours, scale error and friction shall not exceed the tolerances specified in Tables I and II.

4.6.9 High temperature exposure. While at the test temperature the scale error and friction shall not exceed the tolerances specified in Tables I and II. After the indicator has been maintained at room temperature for not less than 4 hours, scale error and friction shall not exceed the tolerances specified in Tables I and II.

4.6.10 Fogging. The indicator shall have been energized and in a controlled ambient of 70° ± 2°C for a minimum of 1 hour. At the end of this period and while at this temperature, an ice cube made of distilled water and of nominal original size, 1 by 1 by 1 inch, with its flat surface 1 by 1 inch, shall be rubbed over a major area of the indicator glass face from 1 to 2 minutes. If the ice cube melts to an unusable size before this test is completed, continue by using a second ice cube. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of internal water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.11 Humidity. Humidity tests shall be as specified in MIL-I-7057.

4.6.12 Electrical zero. Connections between the indicator and the electrical zero transformer shall conform to Figure 4.

4.7 Inspection of preservation, packaging, packing and marking. Inspection of preparation for delivery shall be conducted in accordance with the quality assurance provisions of MIL-STD-794 to insure compliance with the requirements of Section 5.

*5. Preparation for delivery. Unless otherwise specified in the individual contract or order, the following shall apply.

5.1 Preservation and packaging.

5.1.1 Level A. (Reference definition within MIL-STD-794). Indicators shall be packaged in accordance with Method II C of MIL-P-116,one each in a unit container conforming to PPP-B-636, weather-resistant class. Each unit shall be cushioned with MIL-P-26514, Class 2 Polyurethane. Cushioning thickness, density and configuration shall insure that shock is dampened below the maximum permissible nonoperating shock limitations established for the indicator. Completed packs shall be afforded rough handling tests prescribed by MIL-STD-794 and MIL-P-116 to confirm packaging reliability prior to initial shipment.

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